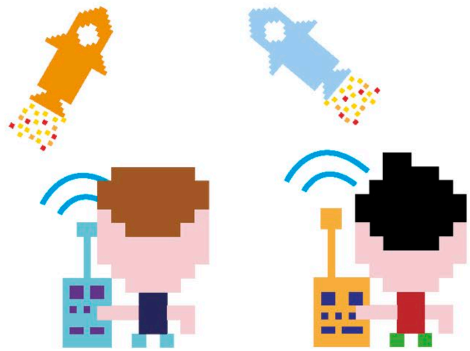
Lists

When you want to store a lot of data, or perhaps the order of the data is important, you may need to use a list. A list can hold many items together and keep them in order. Python gives each item a number that shows its position in the list. You can change the items in the list at any time.



1 Multiple variables
Imagine you're writing a multiplayer game and want to store the names of the players in each team. You could create a variable for each player, which might look like this...

```
>>> rockets_player_1 = 'Rory'
>>> rockets_player_2 = 'Rav'
>>> rockets_player_3 = 'Rachel'
>>> planets_player_1 = 'Peter'
>>> planets_player_2 = 'Pablo'
>>> planets_player_3 = 'Polly'
```

With three players per team, you'd need six variables.

2 Put a list in a variable
...but what if there were six players per team? Managing and updating so many variables would be difficult. It would be better to use a list. To create a list, you surround the items you want to store with square brackets. Try out these lists in the shell.

```
>>> rockets_players = ['Rory', 'Rav',
'Rachel', 'Renata', 'Ryan', 'Ruby']
>>> planets_players = ['Peter', 'Pablo',
'Polly', 'Penny', 'Paula', 'Patrick']
```

The list items must be separated by commas.

This list is stored in the variable **`planets_players`**.

3 Getting items from a list
Once your data is in a list, it's easy to work with. To get an item out of a list, first type the name of the list. Then add the item's position in the list, putting it inside square brackets. Be careful: Python starts counting list items from 0 rather than 1. Now try getting different players' names out of your team lists. The first player is at position 0, while the last player is at position 5.

This line gets the first item in the list, from position 0.

```
>>> rockets_players[0]
'Rory'
>>> planets_players[5]
'Patrick'
```

This line gets the last item in the list, from position 5.

Hit enter/return to retrieve the item.